Problem 3

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Problem 1 *Let f be a function given by $f(x) = \ln(1 + x)$.*

(a) *Find the domain of f. Write your answer in interval notation.*

Explanation. $1 + x > 0 \implies x > -1 \implies$ domain:$(-1, +\infty)$

(b) *Find the vertical asymptotes of f and **EXPLAIN** and justify your answer.*

Explanation. The function $f(x) = \ln(1 + x)$ has a vertical asymptote at $x = -1$ since $\lim_{x\to-1^+} \ln(1 + x) = -\infty$. (There is no Theorem/Test to reference here, since it is checking the definition of a vertical asymptote.) Alternatively, $g(x) = \ln(x)$ has a vertical asymptote at $x = 0$. f is g shifted one unit left so f will have a vertical asymptote at $x = -1$.

(c) *Sketch a graph of f*

Explanation. The graph of $f(x) = \ln(1 + x)$ is show below.

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